FitHub Class Booking System

Business Requirements Document

Field	Entry
Project	FITHUB Class Booking System
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Approved by	Owner, Sponsor
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Table of contents

Revision History....3
Responsibilities (RACI)....3
Executive Summary....4
Overview....4
Background....4
Objectives....4
Requirements summary....4
Proposed strategy....5
Problem Statement....5
Business Objectives....5
Business Requirements....6
Scope....8
Impact of proposed changes....9
Stakeholders and Responsibilities....10
Actors and other systems....11
Business Rules....12
Use cases and Diagrams....12

Non-functional requirements.....17
Test plan..... 18
Implementation and after launch.....19
Constraints ,Assumptions and Dependencies.....19
Risks.....20
Business case and timetable.....21
Approval and Sign-offs..... 22

Revision History

Version	Date	Author	What changed
01	10/15/2026	Dhairya Dholakia	First Draft

Responsibilities (RACI)

R=Responsible (writes it). A= Accountable (approves it, only one). C= Consulted (gives input). I= Informed (keep it up-to-date)

Person (role)	Code	What they do for this BRD
Business Analyst	R	Writes and updates the BRD
Owner (Sponsor)	A	Approves the final version
IT lead	C	Confirms what is technically possible
Front desk lead	C	Confirms how bookings works today

Finance	C	Confirms plan rules and payment details
Data protection / legal	C	Reviews GDPR requirements
Trainer representative	I	Keep informed about the attendee list feature

Executive Summary

3.1 Overview

This project will give FitHub members an online way to view, book and cancel classes on the web, iPhone and Android, and will give front desk staff and trainers the tools they need to manage classes.

3.2 Background

FitHub has 12 gyms and about 18,000 members. Class bookings are taken by phone and at the front desk and recorded in a spreadsheet. This causes weekly double bookings, members who cannot see whether a class is full, and about 30% of booked members not attending. The front desk spends around 2 hours a day on bookings, and trainers have no reliable attendee list or enforced class limit.

3.3 Objectives

Increase class revenue by 10% within 12 months of launch; reduce the no-show rate from 30% to 20% within 6 months; eliminate double bookings within 3 months; cut front desk booking time from 2 hours to 30 minutes a day within 3 months; and have 80% of bookings made online within 3 months.

3.4 Requirements summary

There are 12 business requirements. The main ones are: members can see availability, book and cancel online; classes cannot be booked beyond

capacity; plan limits and cancellation rules are applied automatically; members can join a waiting list; trainers see attendee lists; and personal data is handled in line with GDPR.

3.5 Proposed strategy

Three options were considered: (A) do nothing, (B) improve the spreadsheet process only, and (C) build a web and mobile booking system. Option C is recommended because it is the only option that addresses all five objectives. Options A and B leave the no-show and front desk problems unsolved.

1. Problem Statement

FitHub is a fitness chain with 12 gyms across the country with about 18000 members enrolled . The gym offers Yoga , spin, and HIIT classes . Class bookings are taken by phone and at the front desk and manually recorded in a spreadsheet . This causes weekly double bookings , members who cannot tell whether a class is full or has availability, and about 30% of booked members not attending . The front desk spends around 2 hours a day on bookings, and trainers have no reliable attendee list or enforced class limit to maintain safety.

If nothing changes, FitHub will keep losing class revenue, and members will become more frustrated. The estimated cost of no-shows is TBC with Finance.

2. Business Objectives

ID	Objective	Baseline	Target	Deadline
G1	Increase class revenue	Current class revenue (TBC)	+10%	12 months after launch

G2	Reduce the no-show rate	30%	20%	6 months after launch
G3	Eliminate double bookings	About 1 per week	0	3 months after launch
G4	Reduce front desk time spent on bookings	2 hours a day	30 minutes a day	3 months after launch
G5	Move bookings online	0% online	80% online	3 months after launch

3. Business Requirements

There are 12 business requirements prioritized with MoSCoW :

ID	Requirement	Priority	Source	Objective
BR-01	Members should be able to see the class availability without contacting the gym.	Must	Member survey	G5
BR-02	Members should be able to book or cancel their classes .	Must	Survey, Front Desk	G4,G5
BR-03	The business shall prevent a	Must	Trainer, Frontdesk	G3

	class from being booked beyond its capacity.			
BR-04	Members shall be able to join a waiting list for a full class.	Should	Member survey	G1,G2
BR-05	The business shall apply plan limits and cancellation rules automatically.	Must	Finance	G4
BR-06	Members shall pay by card through the existing payment provider.	Must	Finance	G1
BR-07	Members shall receive booking confirmations and class reminders.	Should	IT lead	G2
BR-08	Trainers shall see the attendee list and class size limit before each class.	Must	Trainer	G3
BR-09	Personal data shall be	Must	IT lead	All

	handled in compliance with GDPR.			
BR-10	The solution shall work on iPhone, Android and common web browsers.	Must	IT lead	G5
BR-11	The solution shall reuse the existing member database.	Must	IT lead	All
BR-12	Frontdesk staff shall be able to view and manage bookings.	Should	Front Desk	G4

4. Scope

The project delivers online class booking for members on the web, iPhone, and Android. In addition to that, the project also delivers the tools that front desk staff and trainers need. Items marked TBC need a client decision

In scope

Members view class availability and book and cancel online via web, Android, and iPhone
Waiting list for full classes
Class credits for Basic and Premium plans , late cancellation rule

Card payments through the existing payment provider
Email and push notifications
Trainer view of attendee list and class limit
Front desk view of bookings

Out of scope

Personal training bookings
Selling new memberships online
Payment methods other than cards
Changes to membership prices or plan types
Door access or check-in hardware
Languages other than English

5. Impact of proposed changes

Process	Today	After the project	Who is affected
Checking class availability	Members call or visit the gym	Members see the availability in the app or website	Members, front desk
Booking a class	By phone or at the front desk, recorded in a spreadsheet	Members book online; the system prevents overbooking	Members, front desk
Cancelling a booking	By phone, rules apply manually	Members cancel online; the	Members , frontdesk,finance

		system applies the 2-hour rule automatically	
Full classes	The member is turned away	The member can join a waiting list and is notified when a spot is available	Members
Attendee list	The trainer asks at the front desk	The trainer sees the attendee list and class limit in the system.	Trainers
Front desk work	About 2 hours a day on bookings	Mostly handles exceptions and members who need assistance	Front desk staff

6. Stakeholders and Responsibilities

Stakeholder	Role in the project	Main need	Power
Owner (Sponsor)	Funds and approves	More class revenue, less front desk effort	High
Front desk lead	Key user	Fewer calls, no double bookings	Medium
Trainers	Key user	Attendee list, hard class limit	Low
Members	End users	See availability, book and cancel	Low

		classes from the phone, and join the waiting list if the class is full	
Finance	Rules and payments	Plan limits, cancellation rule , card payments	Medium
IT lead	Technical owner	Existing database, iPhone/android/ Web browser, notifications	Medium
Data protection/legal	Compliance	GDPR	High
Payment provider	External party	Integration rules	Low

7. Actors and other systems

Type	Name	Function and need
Worker (inside FitHub)	Front desk staff	View and manage bookings , help members who need it
Worker (inside FitHub)	Trainer	View the attendee list and class limit
Worker (inside FitHub)	Admin	Set up classes , schedules and capacity
External party	Member	Views availability ,

		books their classes, cancels their class, joins waiting list
External party	Payment provider	Processes card payments
Other systems	Member database	Existing records of members and plans . the new system reads from it
Other systems	Payment system	Connects bookings to card payments
Other systems	Email and push notification service	Sends confirmations, reminders, and waiting list offers
Other systems	Class spreadsheet	Current tool that will be retired after launch

8. Business Rules

BRU-01: The basic plan includes 5 classes per month. The premium plan is unlimited classes.
BRU-02: Cancelling less than 2 hours before the class costs one class credit.
BRU-03: A class cannot have more bookings than its capacity.
BRU-04: Payments are by card only, through the existing payment provider.

9. Use cases and Diagrams

Figure:1 Booking Workflow (to-be)

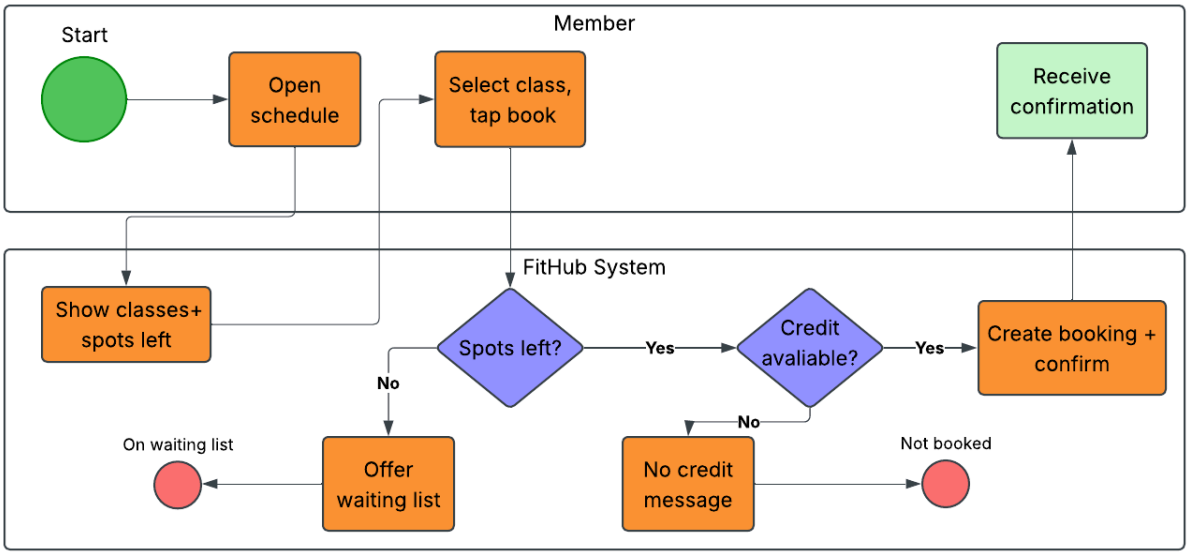
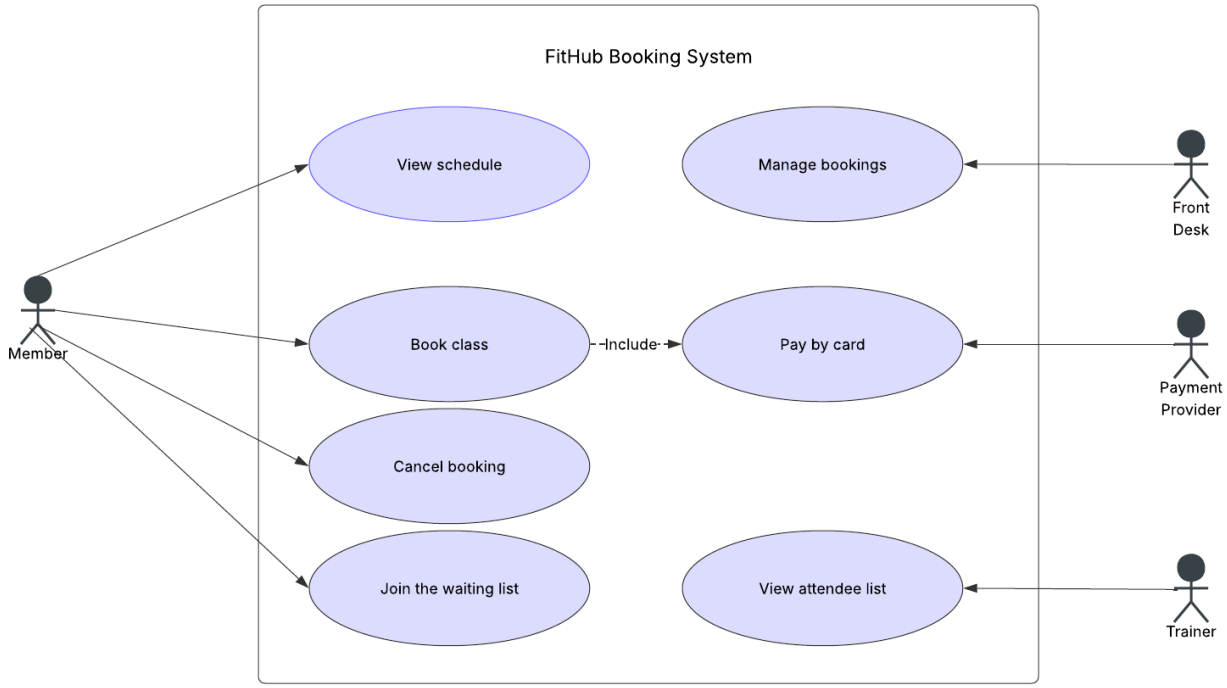


Figure:2 Use case diagram



UC-01 Book a class :

Field	Entry

Name	Book a class
Actor	Member
Goal	Reserve a spot in a class
Before it starts	The member is logged in and has an active membership with the gym.
Result if it works	A booking is made and saved, the spots left go down by one, and the member receives a confirmation message.
Result if it fails	The member's booking is not created, and no credits are used
Business rules used	BRU-01,BRU-03,BRU-04
Requirements	BR-01,BR-02,BR-03,BR-05,BR-06, BR-07

Main steps

1. The member opens the class schedule through the app or the website.
2. The system shows the classes with the number of spots left .
3. The member selects a class and clicks on "Book class".
4. The system checks that spots are left and the member has a credit available.
5. The system creates the booking and reduces the spots left by one.
6. The system shows a confirmation and send an email and push notification to the member.

Other paths:

- 4a. The class is full: the system offers the waiting list (see UC-03).
- 4b. The member has no credits left: the system explains why and no booking is made.
- 4c. Payment is needed and fails: the system tells the member why, creates no booking, and lets the member retry.

- 4d. The member already has a class at that time: the system shows a message and no booking is made.
- 6a. The notification service is down: the booking is still saved and the confirmation is shown on screen.

UC-02 Cancel a booking :

Field	Entry
Name	Cancel a booking
Actor	Member
Goal	Release a booked spot in the system
Before it starts	The member has an active booking.
Result if it works	The spot is released, and the credit is returned or kept according to the cancellation rule .
Result if it fails	The booking stays as it was.
Business rules used	BRU-02
Requirements	BR-02,BR-04,BR-05

Main steps:

1. The member opens My bookings and selects the booking.
2. The member taps Cancel.
3. The system checks how long is left before the class.
4. The system releases the spot and returns the credit.
5. The system shows a confirmation and offers the spot to the first person on the waiting list, if there is one.

Other paths

- 3a. Less than 2 hours remain: the system warns that a credit will be lost, and the member must confirm. If confirmed, the spot is released and the

credit is not returned. If the member declines, the booking stays.

- 3b. The class has already started: the system does not allow cancelling.

UC-03 Join a waiting list :

Field	Entry
Name	Join a waiting list
Actor	Member
Goal	Get a spot in a full class if one opens
Before it starts	The class is full and the member is logged in
Result if it works	The member gets on the waiting list with a position number
Result if it fails	The member is not on the list
Requirements	BR-04, BR-07

Main steps:

1. The member sees that a class is full and taps Join waiting list.
2. The system adds the member and shows their position.
3. When a spot opens, the system offers it to the first member and sends a notification.
4. The member accepts within the time allowed (TBC), and the system creates the booking.

Other paths

1. 4a. The member does not accept in time: the system offers the spot to the next person.
2. 2a. The member leaves the list: the system removes them and updates positions

10. Non-functional requirements

ID	Type	Requirement
NFR-01	Response time	The class schedule should load within 2 seconds for 95% of requests.
NFR-02	Response time	A booking shall be confirmed within 3 seconds for 95% of requests.
NFR-03	Load	The system shall support 2000 users at the time during peak hours (TBC , about 10% of members)
NFR-04	Availability	The system shall be available 99% of the time each month , not counting planned maintenance.
NFR-05	Security	Members shall be able to see and change their own data and bookings .
NFR-06	Security	Personal data shall be encrypted when sent and stored .
NFR-07	Privacy (GDPR)	Members shall give

		consent for notifications and can ask for their data to be deleted.
NFR-08	Compatibility	The system shall work on two latest versions of IOS and Android along with current Chrome, Safari, Edge, and Firefox.
NFR-09	Usability	A new member should be able to book a class within 1 minute without training.
NFR-10	Recovery	Data shall be backed up daily and restored within 4 hours after a failure (TBC with IT lead)

11. Test plan

Approach: Every requirement is tested against its use case steps and rules. Test cases are written from the use cases.
Requirements-based tests: Check each business rule; for example, a full class cannot be booked, a basic member cannot exceed 5 bookings a month, and cancelling less than 2 hours before class costs a credit to the member.
System tests: Check the payment provider, the member database, and notifications working together, and run a load test for peak hours.
User acceptance testing: The front desk lead, a trainer, and a small group of members try the system before launch.

Done when : All the MUST requirements pass and there are no open critical or high-severity defects in the system.

12. Implementation and after launch

Training: Front desk staff receive about 2 hours of training, and trainers about 1 hour before launch. Members receive a short in-app guide and an email on how to navigate the system.
Rollout: Launch in 2 pilot gyms for 4 weeks, fix the problems, then roll out to the other 10 locations. Phone and front desk booking stay available during the rollout.
After launch : Measure G1 to G5 monthly with simple reports and review them at 3, 6, and 12 months after launch. Report results to the owner.

13. Constraints ,Assumptions and Dependencies

Constraints
Budget of about 60,000 euros.
Launch within 6 months
Must comply with GDPR.
Must use the existing payment provider and member database

Assumptions
The existing member database can be connected to the new system (to verify with IT)
The payment provider allows integration for the booking payments (to verify)
Members have a smartphone or a web browser

Dependency
Payment provider support during integration
Availability of the IT lead and front desk for reviews

14. Risks

ID	Risk	Likelihood	Impact	Mitigation
R1	Some members find the app or the site hard to use	Medium	Medium	Simple design, in-gym help, keep phone and FD booking during rollout
R2	The system is slow and fails with 18000 members	Medium	High	Load testing and scalable hosting
R3	Members leave because everything moved online	Low	High	Gradual rollout, communicate the benefits, support for non-app users
R4	Personal data is mishandled	Low	High	Data protection

				review before launch
R5	Payment integration is delayed	Medium	Medium	Early technical check with the provider

15. Business case and timetable

Business case

Item	Estimate
Cost to build	About 60,000 euros (budget)
Yearly running cost (hosting, support, notifications)	TBC with IT lead
Yearly benefit, extra class revenue	10% of currently yearly class revenue (TBC with finance)
Yearly benefit: front desk time saved	About 1.5 hours a day saved x cost per hour x working days (TBC with finance)
Payback time	Cost to build divided by yearly benefit (TBC once the figures above are confirmed)

Timetable

Milestone	Target
BRD approved	End of month 1

Functional requirements and design finished	End of month 2
Build finished	End of month 4
Testing and user acceptance finished	End of month 5
Launch (pilot in 2 gyms, then all 12)	Month 6

16. Approval and Sign-offs

Name	Role	Decision	Date
Owner	Sponsor		
IT lead	Technical owner		